

Feasibility of Training Elite Athletes for Improving their Mental Imagery Ability Using Virtual Reality

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ABSTRACT

The goal of imagery training for athletes is to create realistic images in their minds and to familiarize them with certain procedures, environments, and other aspects related to competition. Traditional imagery training methods use still images or videos, and athletes study the pictures or watch the videos in order to mentally rehearse. However, factors such as distractions and low realism can affect the training quality. In this paper, we present a VR solution and a study that explores our hypotheses that 1) high-fidelity VR systems improve mental imagery skills, and that 2) the presence of elements such as an audience or photographers in the VR environment result in better mental imagery skill improvement.

Index Terms: Human-centered computing—Visualization—Visualization techniques—Treemaps; Human-centered computing—Visualization—Visualization design and evaluation methods

1 INTRODUCTION

Mental imagery is a cognitive process, widely used by athletes as a performance-enhancing technique to facilitate motor actions [3]. Imagery as mental practice can positively affect skilled motor performance, especially when practice conditions are “optimal” [1]. Creating realistic images in the mind is one of the most important criteria that determines the quality of imagery training [2]. However, current imagery-training methods rely on mental training, and are commonly not supported by innovative technology. There is demand for an innovative mental imagery training method to help athletes improve their skills of mental imagery. In this paper, we present a VR-based software solution, explore factors that can affect mental imagery, evaluate the approach using elite athletes, and discuss the positive and negative effects.

For our study, we are working together with a national snow sports organization to support the preparation of Olympic ski and snowboard athletes before elite competitions. The motivation behind the user study is that some athletes reported that they feel nervous and stressed prior to performance. Factors such as expectations and the pre-race environment (audience, photographers, etc.) can affect their performance. Therefore, we built a realistic virtual snow sports environment to assist athletes in the development and subsequent

rehearsal of their pre-performance routine within a VR environment, and explore the impact of different training methods on imagery improvement and affect. For the VR conditions, participants wore a headset and virtually moved themselves from the warm-up tent to the start-gate of a course. Depending on the experimental condition, various types and levels of stimuli were present along the way, including an audience, other athletes and photographers.

Our results show that a crowded VR simulation with a virtual audience, etc. can significantly increase heart rate, which signifies increased emotional arousal. Although the results from validated questionnaires show no significant difference for mental imagery and affect, the results show an ascending trend for athlete arousal from Video to the VR-Crowded condition.

2 METHODS

We recruited seven elite athletes (four male, three female) from the national snow sports organization in our country as participants for the study. These seven participants represent almost all of the Olympic-level snow sports athletes from our country. Participants were aged 15-22 ($M = 19$, $SD = 2.2$).

2.1 Study Design

The main goals of the study was to compare different visual stimuli in a VR-based imagery training method to determine if the conditions affect mental imagery skills, and how the presence or absence of others in the environment affects imagery ability. We adopted a within-subjects design with one independent variable, Training Method, with three levels: Video, VR-Empty, and VR-Crowded (Figure 1). The material for the Video condition was captured and recorded from a session of the VR-Crowded condition, and presented to athletes from a first-person view. The VR-Empty condition followed the same path as VR-Crowded, but we removed other athletes, audience members, photographers, screen display, and cheering sounds from the experience. Therefore, VR-Empty was a less-distracting version of VR-Crowded. The order of exposure to the three conditions was randomized for each participant. Our hypotheses were as follows:

- We predicted that using high-fidelity VR systems to display realistic 3D sport environments (VR-Empty & VR-Crowded) would improve mental imagery skills compared to still-image and video-based training methods (Video), allowing athletes to better prepare for real-world, high-pressure situations.
- We predicted that participants would have better mental imagery skill improvement in an environment when the audience and photographers stand alongside the walking path (VR-Crowded) than in an environment without audience and photographers (VR-Empty). We expected the crowded condition to be more like real competition, since inexperienced participants would have stronger emotional reactions to having people around them.

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Figure 1: VR conditions: a) Warm-up tent of VR-Empty, b) Corridor of VR-Empty, c) Warm-up tent of VR-Crowded, d) Corridor of VR-Crowded

2.2 Measurements

The physiological data, such as Galvanic Skin Response (GSR) and Heart Rate (HR), were measured before and during the user study. We measured mental imagery using the Sport Imagery Ability Questionnaire (SIAQ) from Williams & Cumming [5]. To measure the positive and negative affect after using the different training methods, we used the Positive and Negative Affect Schedule (PANAS) [4].

3 RESULTS

Change in HR was statistically significantly different for the three conditions, $F(2, 12) = 80.56, p < 0.01, \eta_p^2 = 0.57$. Post-hoc analysis with a Bonferroni adjustment revealed that *Change in HR* for VR-Crowded was higher than both Video ($p = 0.02$) and VR-Empty ($p = 0.01$), but there was no difference between Video and VR-Empty ($p = 0.56$). For *Change in GSR*, the differences were not statistically significant, $\chi^2(2) = 2.0, p = 0.36$.

Table 1: Change in Heart Rate (HR) and Galvanic Skin Response (GSR)

	Video	VR-Empty	VR-Crowded
HR: $M(SD)$	4.1 (4.28)	3.2 (3.51)	9.5 (5.01)
GSR: $M(SD)$	0.13 (0.15)	0.23 (0.35)	0.43 (0.27)

Global for SIAQ did not elicit statistically significant differences between the three training methods, $F(2, 12) = 1.26, p = 0.32, \eta_p^2 = 0.17$, and there were also no significant difference between three methods for other four types of imagery ability. (Figure 2).

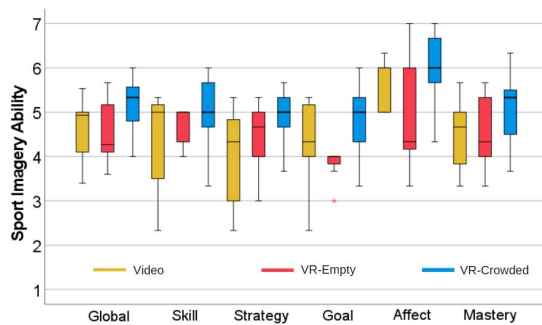


Figure 2: Sport Imagery Ability Questionnaire

The PANAS includes positive and negative parts. The results are shown in Figure 3.

The positive and negative affect did not elicit statistically significant difference respectively, after using any of the three training methods, $F(2, 12) = 0.75, p = 0.49, \eta_p^2 = 0.11$. $F(2, 12) = 0.13, p = 0.88, \eta_p^2 = 0.02$.

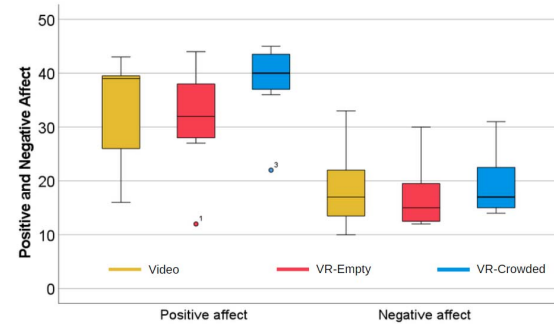


Figure 3: Positive and Negative Affect

4 CONCLUSION

We developed a realistic VR environment to train and improve mental imagery skills of elite snow sports athletes. We designed and implemented a user study to explore the impact of different visual stimuli on imagery ability, and measured their positive and negative affect. The physiological data such as heart rate and galvanic skin response were collected. The results from the athletes show that a high-fidelity VR simulation with the representation of an audience around the athletes can significantly increase heart rate, which indicates an elevation in emotion arousal. Although the results from questionnaires (SIAQ and PANAS) show no significant differences between the three conditions, the results indicate positive trends and feedback using VR for mental imagery training, which partly supports our hypotheses. According to feedback gathered during interviews with the athletes, they reported a clear preference for the innovative training methods (i.e., the VR methods) over traditional video training methods. We look forward to continuing this important journey of helping elite athletes reach their full potential through improved innovative training approaches employing user-centered design approaches we used.

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